



ISI MSQE PEA 2025 PYQ Practice Booklet

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STATSTRIVE

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STATSTRIVE

Question 1

Consider a standard Solow style economy where the aggregate production function is $Y = KN$. Assume no technological progress ($g = 0$) and no population growth ($n = 0$). Let $\bar{K}$ and $\bar{Y}$ be the steady state capital stock and output respectively, $s \in [0, 1]$ the saving rate and $\delta \in [0, 1]$ the depreciation rate. Find the steady-state capital per worker and output per worker.

- (A) $s\delta$ and $(s\delta)^2$
 - (B) s/δ and s
 - (C) s^2/δ^2 and s/δ
 - (D) None of the above
-

Question 2

Continue with Q1. Assume $\delta = 0.1$. At the steady state, consumption per worker and the saving rate that maximises consumption per worker are given by, respectively:

- (A) $s(1 - s)$ and $s = 0.5$
 - (B) $(s/\delta)^2$ and $s = 0.4$
 - (C) s/δ^2 and $s = 0$
 - (D) $\delta/(1 - s)$ and $s = 1$
-

Question 3

Wage setting: $W = P(1 - u)$. Production: $Y = N$. Markup $\mu = 20\%$. Labour force $L = 60$. Find the natural rate u_n and natural output Y_n .

- (A) 16.667% and 50
 - (B) 23.5% and 15.5
 - (C) 20% and 15.5
 - (D) 20% and 20
-

Question 4

Open economy, fixed exchange rate = 1. $IM = 0.3Y$, $X = 0.3Y^*$. The foreign country has the same structure. Find the domestic multiplier.

- (A) 1

- (B) 2
 - (C) 0.4
 - (D) 0.8
-

Question 5

Continue with Q4. Target $Y = 250$, foreign G^* unchanged. Required increase in G ?

- (A) 112.4
 - (B) 96
 - (C) 38.8
 - (D) 86
-

Question 6

$Y = A \min\{K, L\}$, $A > 0$. With $W = \partial Y / \partial L$ and $R = \partial Y / \partial K$:

- (A) R and W same for every k .
 - (B) Same for $k \leq 1$ but not for $k > 1$.
 - (C) Same for $k > 1$ but not for $k \leq 1$.
 - (D) None of the above.
-

Question 7

$\frac{d}{dt} [e^{-\rho t} U'(C)] = e^{-\rho t} U'(C) r$. When is $\dot{c} > 0$?

- (A) $r < \rho$
 - (B) $r > \rho$
 - (C) $r = \rho$
 - (D) Never.
-

Question 8

Income stream $\{y_h, y_l, y_h, y_l, \dots\}$, discount factor $\beta \in (0, 1)$. Find constant $\bar{c}$ with equal present value.

- (A) $\beta(y_h + y_l)/(1 - \beta)$
 - (B) $\beta^2(y_h + y_l)/(1 - \beta)$
 - (C) $\beta^2(y_h + y_l)(1 - \beta)$
 - (D) $(y_h + \beta y_l)/(1 + \beta)$
-

Question 9

Out of 20 questions, average correct = 11.7, average wrong = 5.3. Has the TA made a mistake?

- (A) Yes
 - (B) No
 - (C) Maybe
 - (D) Insufficient information
-

Question 10

Under the null $p = 1/2$, compare $P(15 \text{ of } 20 \text{ smokers die first})$ vs. $P(12 \text{ of } 20)$.

- (A) $= 1$
 - (B) < 1
 - (C) > 1
 - (D) Insufficient info.
-

Question 11

$\hat{\beta}_0 = 5$, $\bar{y} = 65$, $\bar{x} = 90$. Predict y for $x = 75$.

- (A) 55
 - (B) 75
 - (C) 40
 - (D) 65
-

Question 12

Two fair coins. A : head on first coin. C : head on second coin. D : coins match. G : two heads. Which statement is false?

- (A) C and D independent.
 - (B) A and G independent.
 - (C) A and D independent.
 - (D) A and C independent.
-

Question 13

Bowl: 3 chips = Re. 1, 2 chips = Rs. 4. Draw 2 without replacement. Expected sum?

- (A) < 2
 - (B) 3
 - (C) (3, 4)
 - (D) (4, 5)
-

Question 14

Two samples; R^2 of first $> R^2$ of second. Which follows?

- (A) Coefficient on X greater in first.
 - (B) |coefficient| on X greater in first.
 - (C) Variance of Y smaller in first.
 - (D) None of the above.
-

Question 15

$sd(x) = 2$, $sd(y) = 4$. The OLS slope of Y on X equals:

- (A) $8r$
 - (B) $2r$
 - (C) $r/2$
 - (D) None.
-

Question 16

$f(x) = 4x^3 - 6x^2 - 72x$. Where is f increasing / decreasing?

- (A) Increasing everywhere.
 - (B) Decreasing everywhere.
 - (C) Increasing in $(-\infty, -2)$, decreasing in $(-2, \infty)$.
 - (D) Increasing in $(-\infty, -2)$ and $(3, \infty)$, decreasing in $(-2, 3)$.
-

Question 17

Find $\sum_{k=1}^{2024} \frac{1}{k(k+1)}$.

- (A) 1
 - (B) 2025/2026
 - (C) 2024/2025
 - (D) 2025/2024
-

Question 18

Find non-negative x, y with $x + y = 16$ minimising $x^3 + y^3$.

- (A) $x = y = 8$
 - (B) $x = 15, y = 1$
 - (C) $x = 12, y = 4$
 - (D) None.
-

Question 19

$\int_1^2 \log x \, dx$.

- (A) $\log 4$
 - (B) $\log 4 - 1$
 - (C) $\log 2 - 1$
 - (D) None.
-

Question 20

Find $6^{2025} \bmod 25$.

- (A) 0
 - (B) 1
 - (C) 24
 - (D) 6
-

Question 21

$f : \mathbb{N}_+ \rightarrow \mathbb{N}_+$, $f(x+y) = f(x)f(y)$, $f(1) = 2$. Find $2 + \sum_{n=1}^{2025} f(n)$.

- (A) 2^{2025}
 - (B) 2^{2026}
 - (C) 3^{2024}
 - (D) None.
-

Question 22

Two demand groups. Both stop at $p = 5$. Per Re. price decrease: student demand +5, professor demand +10. $MC = 1$. Who is charged more?

- (A) Professors
 - (B) Students
 - (C) Same
 - (D) Insufficient info.
-

Question 23

Bharat: linear demand, zero above $p = 5$. Chinmayi: linear demand, zero above $p = 10$. Market demand kink and flatness?

- (A) Kink at Rs. 5, flatter to right.
 - (B) Kink at Rs. 5, flatter to left.
 - (C) Kink at Rs. 15, flatter to right.
 - (D) Kink at Rs. 15, flatter to left.
-

Question 24

Income = 200. $p_B = 5$, $p_Y = 1$. $U = 4B + 2Y$. Plus a Rs. 50 gift card usable only on books.

- (A) 10 books, 200 other goods.
 - (B) 0 books, 200 other goods.
 - (C) 20 books, 150 other goods.
 - (D) 40 books, 0 other goods.
-

Question 25

$MC = 25$. 5 “high” people have $MB_H = 100 - 0.5Q$, 10 “low” people have $MB_L = 50 - 0.25Q$. Max number of trees voluntarily provided with cost-sharing?

- (A) 0
 - (B) 100
 - (C) 150
 - (D) 195
-

Question 26

$x \succ y$ iff (i) $x_a + x_b > y_a + y_b$, or (ii) $x_a + x_b = y_a + y_b$ and $x_a > y_a$. Which statements are true?

I. $d_a = w/p_a$ if $p_a < p_b$. **II.** $d_a = w/p_a$ if $p_a \leq p_b$. **III.** $d_a = 0$ if $p_b \leq p_a$. **IV.** $d_b = w/p_b$ if $p_b < p_a$. **V.** $d_b = 0$ if $p_b < p_a$.

- (A) I and III
 - (B) II and III
 - (C) II and IV
 - (D) I and V
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Question 27

Same setting as Q26. Are d_a, d_b continuous in (p_a, p_b) ?

- (A) Both continuous.
 - (B) d_a continuous, d_b not.
 - (C) d_a not continuous, d_b continuous.
 - (D) Neither continuous.
-

Question 28

$U(x_1, x_2) = 4x_1^2 + x_2^2$. Solve $\max U$ subject to $p_1x_1 + p_2x_2 = w$. Let $X(p_1, p_2, w)$ be the solution set. Which is true?

- (A) $X(2, 1, 10)$ not convex.
 - (B) $X(6, 3, 30)$ convex.
 - (C) $X(1, 1, 10)$ not convex.
 - (D) $X(4, 2, 20)$ convex.
-

Question 29

$U(x, y) = \frac{x}{1+x} + y$, $p_x = p_y$. Then:

- (A) $x > 0, y = 0$.
 - (B) $x = 0, y > 0$.
 - (C) $x > 0, y > 0$.
 - (D) $x = y = 0$.
-

Question 30

Two consumers, same preferences, same endowments. Strictly convex preferences. Then the economy has:

- (A) No Pareto-efficient allocation.
 - (B) Multiple competitive equilibrium allocations.
 - (C) Exactly two Pareto-efficient allocations.
 - (D) Only one competitive equilibrium allocation.
-